



A Machine Learning Approach for Micro-Credit Scoring

based on the paper of Apostolos Ampountolas et al.

Camille Rouan & Sacha Sagratella,

December 2025

Contents

1	Introduction	2
2	Simulation	2
3	SMOTE-NC	2
4	Adjusting the decision threshold (α)	3
5	Results	4
	References	6
6	Appendix	7

1 Introduction

This empirical project based on the paper of Apostolos Ampountolas et al. aims to recreate the main classification methods used in the original paper, Random Forest, AdaBoost and XGBoost with the oversampling technique SMOTE-NC. Then, compare the models performance from the baselines (with unbalanced data), after balancing (using SMOTE-NC) and with a third scenario: adjusting the decision probability of the low frequency alternative (here the risk class "average or good"). The original paper "compares various machine learning algorithms on real micro-lending data to test their efficacy at classifying borrowers into various credit categories" (Apostolos et. al, 2021). For feasibility we simulated data and made some slight variation from the original work.

2 Simulation

We begin by simulating a subset of the variables used in the paper. The goal is to reproduce the marginal distributions of three variables : *age*, *log amount of loan*, and *annualized interest rate*.

Based on the descriptive statistics reported in the paper, both *age* and *log amount of loan* have skewness and kurtosis values close to zero, which is consistent with an approximately normal distribution. Then, we simulated these two variables using Normal distribution with the mean and the standard deviation reported in the original database. In addition, we arbitrary decided to range age between 18 and 80 to ensure realistic data for credit. However, the *annualized interest rate* shows extremely high skewness and kurtosis, indicating asymmetric and heavy tailed distribution. We modeled the variable using Log-Normal distribution for reasonable approximation.

Last but not least step for simulation was to create the main variable *risk class*. In the paper the categorical variable was presented with three classes, for simplicity we assume a slight change from initial work. Indeed, we merged *average* and *good* categories into a single group, resulting in two classes: *poor* and *average or good*. The class proportion are the same as in the paper, 83.72% of observations belonging to the *poor* risk class.

3 SMOTE-NC

The dataset is strongly unbalanced, this can affect the performance of machine learning algorithms. To address this issue, the authors applied *SMOTE-NC*, an oversampling technique for unbalanced datasets. This extension of *SMOTE* can handle both numerical and categorical features. Chawla et al. (2002) [2] describe SMOTE as a method that over-samples the minority class "by taking each minority class sample

and introducing synthetic examples along the line segments joining any/all of the k minority class nearest neighbors”. We use $k = 5$ nearest neighbors. Figure 1 illustrates the distribution of the two classes before and after applying SMOTE-NC, showing that the oversampling successfully balances the dataset.

After obtaining the balanced dataset, we split it into a training and a testing set (80% / 20%). We then evaluate three supervised learning algorithms commonly used in credit-risk modeling: Random Forest, Adaboost, and XGBoost. These algorithms were the three with the best accuracy from the list of seven in the original work.

For each method, the hyperparameters were set according to those reported in the paper. For each model, we compute four performance metrics: accuracy, precision, recall, and f1-score, with a particular emphasis on the minority class (*average or good*). In addition we plot the ROC curve and compute the AUC each time.

4 Adjusting the decision threshold (α)

In a second time, we decided to adjust the decision probability of a class. Indeed, when one of the alternatives has a very low probability, the classifier tends to under predict this alternative. Usually, in binary classification, models assign an observation to the positive class if the predicted probabilities exceeds a default threshold 0.5. However, in imbalanced datasets, this default choice may not be optimal. The idea is to choose a threshold, say α , optimal and lower than 0.5 e.g.

$$\hat{y}_i(\alpha) = \begin{cases} \text{average or good} & \text{if } \widehat{\Pr}(Y = \text{average or good} \mid X = x_i) > \alpha, \\ \text{poor} & \text{otherwise.} \end{cases}$$

To choose the right α , for each model (Random Forest, AdaBoost and XGBoost), we computed model probabilities and computed different metrics of performance (accuracy, precision, recall and f1-score) across a range of possible α ’s thresholds, from 0.1 to 0.9. The optimal threshold was then selected with an arbitrary choice between maximum accuracy and maximum f1-score on the minority class. Indeed, we cannot choose the threshold with the maximum accuracy, because in this case the f1-score is very low, meaning that the model doesn’t succeed to identify well the minority class. On the contrary, a maximum f1-score is often coming with a low accuracy, meaning that the model predicts poorly both classes. By adjusting α , we can explicitly control this trade-off. This is particularly relevant in credit scoring, where the cost of misclassifying a high-quality borrower as *poor* can differ significantly from the opposite.

The adjusted- α models were then compared to the baseline models trained on imbalanced data and to the

SMOTE-NC-balanced models.

5 Results

The results of our analysis are presented in the Appendix. Descriptive statistics are presented in Table 1. Figure 1 illustrates the data balancing effect of SMOTE-NC on the risk class distribution. Figure 2 presents the ROC curves with their associated AUC for each classification model. Table 2 summarizes the different model performances depending on the adjusted α threshold. Finally, Table 3 presents a complete comparison of all methods used (baseline, SMOTE-NC and adjusted α) across different metrics: accuracy, recall, precision and f1-score.

According to the descriptive statistics in Table 1, we observe that the simulated data approximates the structure of the original dataset. The goal was to reproduce the main distributional properties to allow meaningful model comparison.

Table 2 illustrates how classification thresholds influence the balance between correctly identifying minority-class observations (recall) and maintaining overall correctness (accuracy). For example, for AdaBoost, the maximum f1-score appears when $\alpha = 0.1$ with a value of 0.3041, but, in the same time the accuracy is very poor, only 0.1793, making the classifier poorly optimal despite the improvement in minority-class detection. By taking $\alpha = 0.3$, we have a f1-score equals to 0.2525, which is slightly lower, with an accuracy of 0.4086, which is more than twice higher. We could also choose $\alpha = 0.4$ to have a f1-score of 0.2081 and an accuracy of 0.6889, but we arbitrary choose to favor an higher f1-score with $\alpha = 0.3$. Similarly, we keep $\alpha = 0.2$ for Random Forest and $\alpha = 0.1$ for XGBoost.

The results presented in Table 3 are very interesting. For each model, the baseline scenario gives high accuracy and poor other metrics. This was expected, indeed, with an unbalanced dataset of proportion 87% / 13%, if the model predict only the majority class for all observations, the accuracy will be close to 80%, but it will fail to detect the minority class. For Random Forest, we have accuracy = 0.816 while precision = 0.167, recall = 0.006 and f1-score = 0.012. In comparison, when we adjust the α the metrics change. Indeed, we lose in accuracy but gain in other metrics, in particular f1-score. We notice that for XGBoost, the accuracy decreases from 0.790 in the baseline to 0.631 while the f1-score increases from 0.098 to 0.209. Finally, SMOTE-NC clearly gives the best results in term of all metrics. For each model there is a slight loss in terms of accuracy but largely offset by large increase of precision, recall and f1-score. For instance, with AdaBoost we observe all the metrics above 0.76 meaning that the model identifies creditworthy borrowers effectively while keeping misclassifications errors at a low level, essential in a (micro) credit-scoring context. Moreover, if we look at the ROC Curves and AUC of the 3 ensemble classifiers with SMOTE-NC technique

used (Figure 2), we can see the three curves are oriented towards top-left corner, which means the 3 models seem to well categorize the different borrowers. Indeed, the AUC's are equal to 0.866, 0.845 and 0.75 for the AdaBoost, Random Forest and XGBoost models respectively. Even though XGBoost has a lower AUC, the three models remain above the commonly used threshold of 0.7 to consider them as well classifiers.

As the second technique used to treat the imbalanced data is an adjusted decision threshold, we only focus on one single threshold and therefore cannot plot the roc curves and compute the AUC's. Therefore, we analyze the confusion matrix of each model. We observe that most of the time, models classify well the poor category, but fail to recognize average or good borrowers, and mismatch the two classes. For example, in the case of the Random Forest model, 519 borrowers were predicted as poor risk class and were actually well classified, 45 were well predicted as average or good, 204 were predicted as average or good while actually being poor and 113 were misclassified as poor risk class.

References

- [1] Apostolos Ampountolas et al. (2021). A Machine Learning Approach for Micro-Credit Scoring. *Risks* 9: 50. <https://doi.org/10.3390/risks9030050>
- [2] N. V. Chawla et al. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research* 16 321–357. <https://doi.org/10.1613/jair.953>

6 Appendix

Table 1: Descriptive statistics of the simulated variables

Variables	Mean	SD	1st Quartile	Median	3rd Quartile	Skewness	Kurtosis
age	41.0996	12.6900	31.5323	40.0833	49.8142	0.3727	-0.3705
log_amount	7.0950	0.7281	6.5936	7.0823	7.6177	0.0267	-0.1942
annualized_interest_rate	3.1156	17.7561	0.0896	0.3343	1.3659	19.9056	544.2631

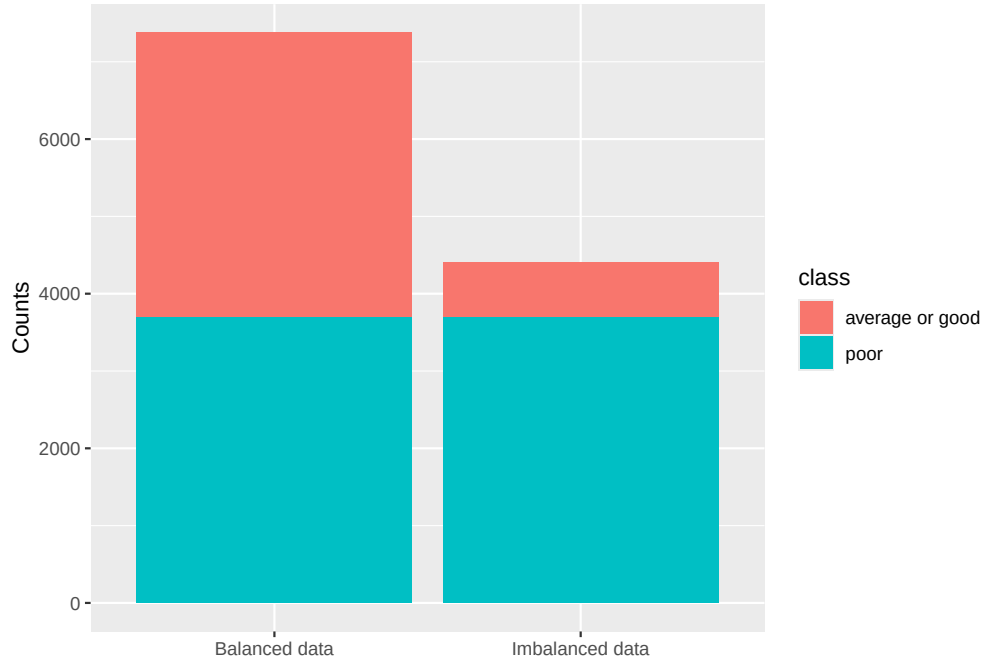


Figure 1: Data balancing using SMOTE-NC

Table 2: Performance metrics for different values of α for Random Forest, AdaBoost and XGBoost

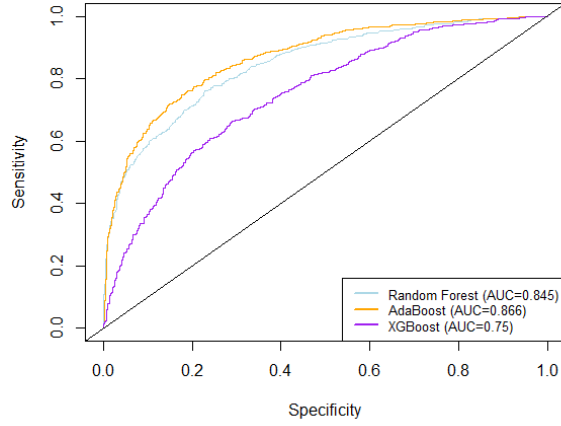
α	Random Forest			AdaBoost			XGBoost		
	Accuracy	Recall	f1-score	Accuracy	Recall	f1-score	Accuracy	Recall	f1-score
0.1	0.3723	0.6329	0.2656	0.1793	1.0000	0.3041	0.6311	0.2722	0.2093
0.2	0.6402	0.2848	0.2211	0.1952	0.9623	0.3000	0.7139	0.1392	0.1487
0.3	0.7667	0.1076	0.1417	0.4086	0.5569	0.2525	0.7480	0.1076	0.1328
0.4	0.8082	0.0253	0.0452	0.6889	0.2278	0.2081	0.7696	0.0823	0.1134
0.5	0.8161	0.0063	0.0122	0.7911	0.0443	0.0707	0.7900	0.0633	0.0976
0.6	0.8207	0.0000	—	0.8138	0.0063	0.0120	0.7968	0.0444	0.0725
0.7	0.8207	0.0000	—	0.8207	0.0000	—	0.8059	0.0199	0.0339
0.8	0.8207	0.0000	—	0.8207	0.0000	—	0.8138	0.0127	0.0238
0.9	0.8207	0.0000	—	0.8207	0.0000	—	0.8207	0.0063	0.0125

Note: Accuracy is computed over all classes, whereas Recall and f1-score refer exclusively to the minority class (“average or good”).

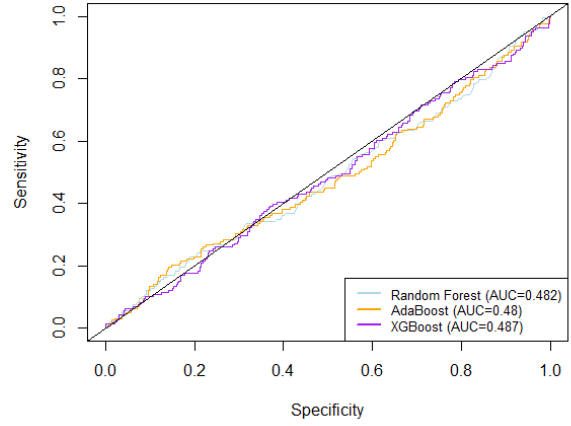
Table 3: Comparison of classification methods

Metric	Random Forest			AdaBoost			XGBoost		
	Standard	SMOTENC	$\alpha=0.2$	Standard	SMOTENC	$\alpha=0.3$	Standard	SMOTENC	$\alpha=0.1$
Accuracy	0.816	0.757	0.640	0.791	0.780	0.409	0.790	0.678	0.631
Precision	0.167	0.738	0.181	0.175	0.760	0.163	0.213	0.670	0.170
Recall	0.006	0.791	0.285	0.044	0.816	0.557	0.063	0.691	0.272
F1-score	0.012	0.763	0.221	0.071	0.787	0.253	0.098	0.681	0.209

Note: Accuracy is computed over all classes, whereas Precision, Recall and f1-score refer exclusively to the minority class (“average or good”).



(a) ROC Curves (SMOTE-NC)



(b) ROC Curves (Imbalanced data)

Figure 2: Comparison of ROC curves for SMOTE-NC and alpha-adjusted methods